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SECTOR AT A GLANCE | 08

Medical Education and Research

Health Sector in India

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Medical Education and Research

Health Sector in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Medical education and research in India serve as the foundation for training individuals with a passion for healthcare to become physicians and surgeons or researchers, thus playing a pivotal role in the nation's overall healthcare ecosystem. Medical education mostly comprises of Allopathy, Dental, Nursing, Paramedical, Physiotherapy and AYUSH. Although India has made notable progress in producing healthcare professionals, a significant concern still revolves around the doctor-to-patient ratio. To address this imbalance and enhance the quality of medical education, the Indian government has initiated strategic efforts, including curriculum reforms and innovative medical technology workshops. The National Medical Commission and Department of Health Research, has been at the forefront of these transformative changes.¹

In this document, we will look into the state of medical education and research in India, exploring its challenges, developments, and the collective efforts aimed at advancing healthcare education and delivery.

A. Background of Medical Education in India



Presently, Medical education in India is overseen and regulated by various statutory bodies that play crucial role in ensuring the quality, standards, and ethical practices within the field of medical studies. These bodies establish guidelines, accredit institutions, define curriculum, and set standards for medical education and practice. However, the foundation of medical education was laid long before, dating back to the era of British colonial rule who felt the need to teach medicine through the establishment of various medical institutions across the country.

1835

Calcutta Medical College - First institution dedicated to providing structured education in Western medicine.²



1835

Madras Medical College - The Madras Medical College (MMC) was founded in the 2nd of February, 1835 to grow the importance of medical education not only in Calcutta but also in other parts of the country.²



1845

Grant Medical College - Grant Medical College officially opened its doors in May 1845 under Sir Robert Grant's governorship.²



1928

All India Medical Association, Kolkata - Established during the 5th All India Medical Conference in Calcutta with a mission to promote advancement of medical and related sciences.²



1933

Formation of Medical Council of India (MCI) - The enactment of the Indian Medical Council Act (IMC Act) in 1933 paved the way for the formal establishment of the Medical Council of India (MCI) as a recognized legal entity to provide framework for organizing and regulating medical education and practice within the country.^{3,4}



1956

Indian Medical Council (IMC) Act Amendment, 1956, 1964, 1993, 2001 - IMC Act was amended over the years to broaden the scope and include several things such as reconstitution of MCI, maintenance of Medical Register for India, regulations on conducting screening test, etc.⁴



2019

Establishment of National Medical Commission (NMC) - A new act known as the National Medical Commission Act, 2019, was enacted, leading to the establishment of the National Medical Commission (NMC) in place of the former Medical Council of India⁴



1. Kandi V. (2022). Medical Education and Research in India: A Teacher's Perspective. Cureus, 14(5), e24680.

2. Reddy A et al. (n.d.). History Revisited: The origins of medical education in India.

3. Ministry of Health and Family Welfare. (2023). Medical Education. Retrieved 21 September 2023.

4. Ministry of Health and Family Welfare. (2016). Medical Education Policy and Medical Education.

National Medical Commission⁵

The National Medical Commission (NMC) was established through the enactment of the National Medical Commission Act, 2019, which became effective on September 25, 2020, as announced in a gazette notification dated September 24, 2020.

Mission and Vision:

- Improve access to quality and affordable medical education.
- Ensure availability of adequate and high-quality medical professionals in all parts of the country.
- Promote equitable and universal healthcare that encourages community health perspective and makes services of medical professionals accessible to all the citizens.
- Encourage medical professionals to adopt latest medical research in their work and to contribute to research.
- Objectively assess medical institutions periodically in a transparent manner.
- Maintain a medical register for India.
- Enforce high ethical standards in all aspects of medical services.
- Have an effective grievance redressal mechanism.

Functions of National Medical Commission⁵

- Lay down policies for maintaining a high quality and high standards in medical education and make necessary regulations in this behalf.
- Lay down policies for regulating medical institutions, medical researches and medical professionals and make necessary regulations in this behalf.
- Assess the requirements in healthcare, including human resources for health and healthcare infrastructure and develop a road map for meeting such requirements.
- Promote, co-ordinate and frame guidelines and lay down policies by making necessary regulations for the proper functioning of the commission, the autonomous boards and the state medical councils.
- Ensure co-ordination among the autonomous boards.
- Take such measures, as may be necessary, to ensure compliance by the state medical councils of the guidelines framed and regulations made under this act for their effective functioning under this act.
- Exercise appellate jurisdiction with respect to the decisions of the autonomous boards.
- Lay down policies and codes to ensure observance of professional ethics in medical profession and to promote ethical conduct during the provision of care by medical practitioners.
- Frame guidelines for determination of fees and all other charges in respect of fifty per cent. Of seats in private medical institutions and deemed to be universities which are governed under the provisions of this act.

5. [NMC. \(2020\). National Medical Commission. \(Retrieved on 27 October 2023\).](#)



Some notable milestones across various branches of Medical Education

Medical



- Government of India (GoI) has implemented diverse measures to enhance the number of medical colleges from 387 colleges in 2014 to 660 medical colleges in 2023.⁶
- National Medical Commission plans to introduce the National Exit Test (NExT) in May 2024, replacing the final MBBS examinations.⁷

Dental



- Currently there are 329 dental colleges with 60 in the government sector and 269 in the private sector.⁸
- Government has initiated a Draft National Dental Commission Bill to replace with former Dentist's Act.⁸

Nursing



- In 2023, the Cabinet Committee on Economic Affairs granted approval for the establishment of 157 additional nursing colleges in co-location with existing medical colleges and allocated a budget of INR 1,570 Crore for this establishment.⁹
- The primary aim of these nursing colleges is to generate an estimated annual output of about 15,700 nursing graduates.⁹

Paramedical



- Government of India introduced National Commission for Allied and Healthcare Professions (NCAHP) Act 2021 to regulate allied and healthcare professionals.¹⁰
- Government is planning to offer more short-term courses for paramedics and assign them a larger role in public health.

Physiotherapy



- Physiotherapists in India are regulated by the provisions of Rehabilitation Council of India (RCI) Act and its Rules and Regulations and not by any other Act or law in India.¹¹
- Physiotherapy has gained recognition and importance in India; however, its full potential is yet to be well-utilised Nationwide.

AYUSH



- In 2022-23, there were 13,274 UG candidates and 5,566 PG candidates in the Ayurveda, Siddha, Unani and Homeopathy streams.¹²
- 260 Students from 32 countries are studying AYUSH in India under the fellowship scheme.¹²

6. Amrit kaal. (n.d.). *Ministry of Information and Broadcasting*.

7. NMC. (2022). *Comments on draft regulations regarding National Exit Test (NExT)*.

8. Dental Council of India. (n.d.). *College List*. Retrieved on 3rd October 2023.

9. Services Export Promotion Council. (2023). *Ministry of Commerce and Industry*.

10. IndiaCode. (2021). *The National Commission for Allied and Healthcare Professions Act, 2021*.

11. IMA. (2023). *Physiotherapists cannot use the prefix "dr"*. Indian Medical Association. Retrieved on 8th October 2023.

12. Ministry of Ayush. (n.d.). *Education*. Retrieved on 3rd October 2023.

B. Background of Medical Research in India¹³



Formation of Department of Health Research (DHR)

The Department of Health Research (DHR) was established as a distinct entity within the Ministry of Health and Family Welfare through a notable amendment to the Government of India (Allocation of Business) Rules, 1961, on **September 17, 2007**. This significant development marked the formal recognition of the need for a specialized department to cater to research and related activities in the domain of public health within the Indian government. It began its operations in **November 2008** upon the appointment of its inaugural secretary.



Aim: Goal of DHR is to bring modern health technologies to the people through research and innovations related to diagnosis, treatment methods and vaccines for prevention; to translate them into products and processes and, in synergy with concerned organisations, introduce these innovations into public health system.



Functions of Department of Health Research

- ▶ Encouraging and organising fundamental, applied, and clinical research, including clinical trials and operational studies, in medical, health, biomedical fields, and the medical profession and education. This involves enhancing infrastructure, expertise, and skills in cutting-edge areas and effectively managing associated information.
- ▶ Promoting and providing guidance on research governance issues, including ethical issues in medical and health research.
- ▶ Coordinating across various sectors and fostering partnerships between the public and private sectors in research areas related to medicine, biomedicine, and health.
- ▶ Providing advanced training in research domains related to medicine and health, including granting fellowships for such training within India and internationally.
- ▶ Facilitating international collaboration in medical and health research, including participation in international conferences related to these domains in India and abroad.
- ▶ Providing technical assistance for handling epidemics and natural disasters.
- ▶ Investigation of outbreaks due to new and exotic agents and development of tools for prevention.
- ▶ Addressing matters concerning scientific societies, associations, as well as charitable and religious endowments in the fields of medicine and health research.
- ▶ Coordinating efforts between organisations and institutes under both the Central and State Governments in areas relevant to the department's responsibilities, and advancing specialized studies in medicine and health.
- ▶ Overseeing and monitoring the functions of the Indian Council of Medical Research (ICMR).

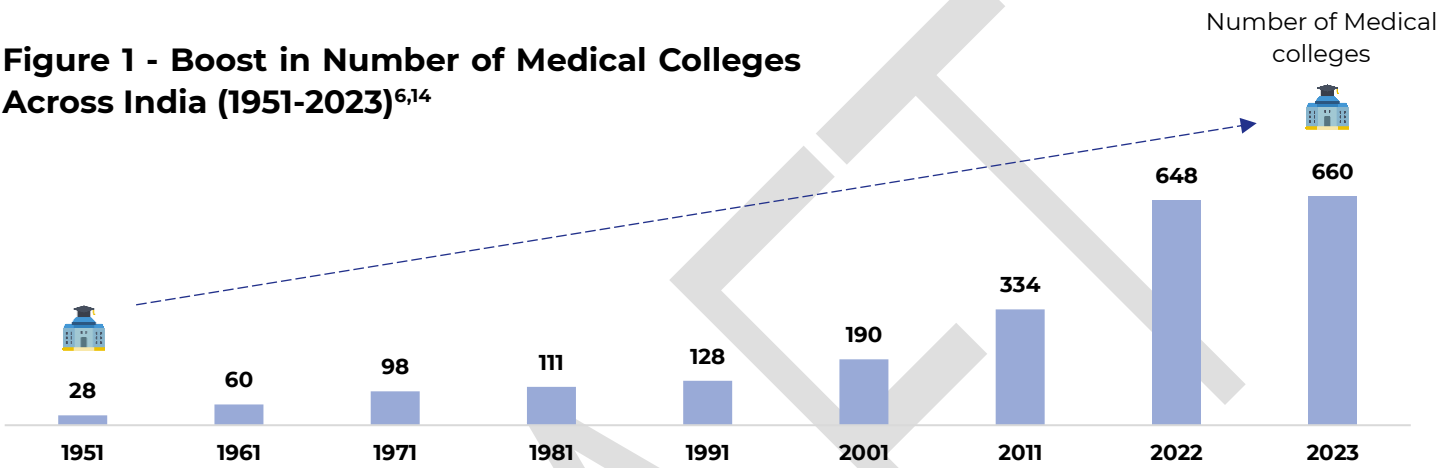
¹³. Department of Health Research. (2023). Annual Report. 2022-2023. Ministry of Health and Family Welfare.

Trends in Medical Education and Research

A. Medical Education

There has been a **remarkable increase** in the number of medical colleges from **1951 to 2023**, demonstrating rapid expansion of medical education in India and increasing emphasis on bolstering healthcare infrastructure.

Figure 1 - Boost in Number of Medical Colleges Across India (1951-2023)^{6,14}



Present Scenario (as of 2023)

Number of AIIMS



From 2014 to 2023, number of All India Institute of Medical Sciences (AIIMS) institutes has experienced threefold growth, escalating from seven to 22.⁶

Number of MBBS Seats



Number of MBBS seats have soared by 97% since 2014, currently standing at a significant count of 1,01,043 (in 2023)¹⁵.

Number of PG Seats



Since 2014, the number of postgraduate (PG) seats has doubled, going from 31,185 to 65,335 in 2023¹⁵.

Key Highlights:

- By enacting the National Medical Council Act in 2019, the Medical Council of India was replaced with the newly formed **National Medical Commission** in the year 2020.¹⁶
- The National Medical Commission is planning to introduce the **National Exit Test (NExT)**, to replace the university-level final MBBS examination commencing from the next year May 2024 to standardize the assessment process for medical graduates across the country.⁷
- Since 2014, the GoI has approved **157 new medical colleges** attached to existing district/referral hospitals.¹⁶
- **INR 6,500 crores** for 2023-24 has been allocated out of Pradhan Mantri Swasthya Suraksha Nidhi (PMSSN) to Human Resources for Health and Medical Education for establishing new medical colleges (upgrading district hospitals), nursing services, pharmacy colleges, etc.¹⁷

14. PIB. (2023). *Taking Medical Education to All Corners of India*.

15. PIB. (2023). *Revolutionizing Medical Education in India*.

16. MoHFW. (2020). *Governance Reforms in Medical Education (2014-2020)*.

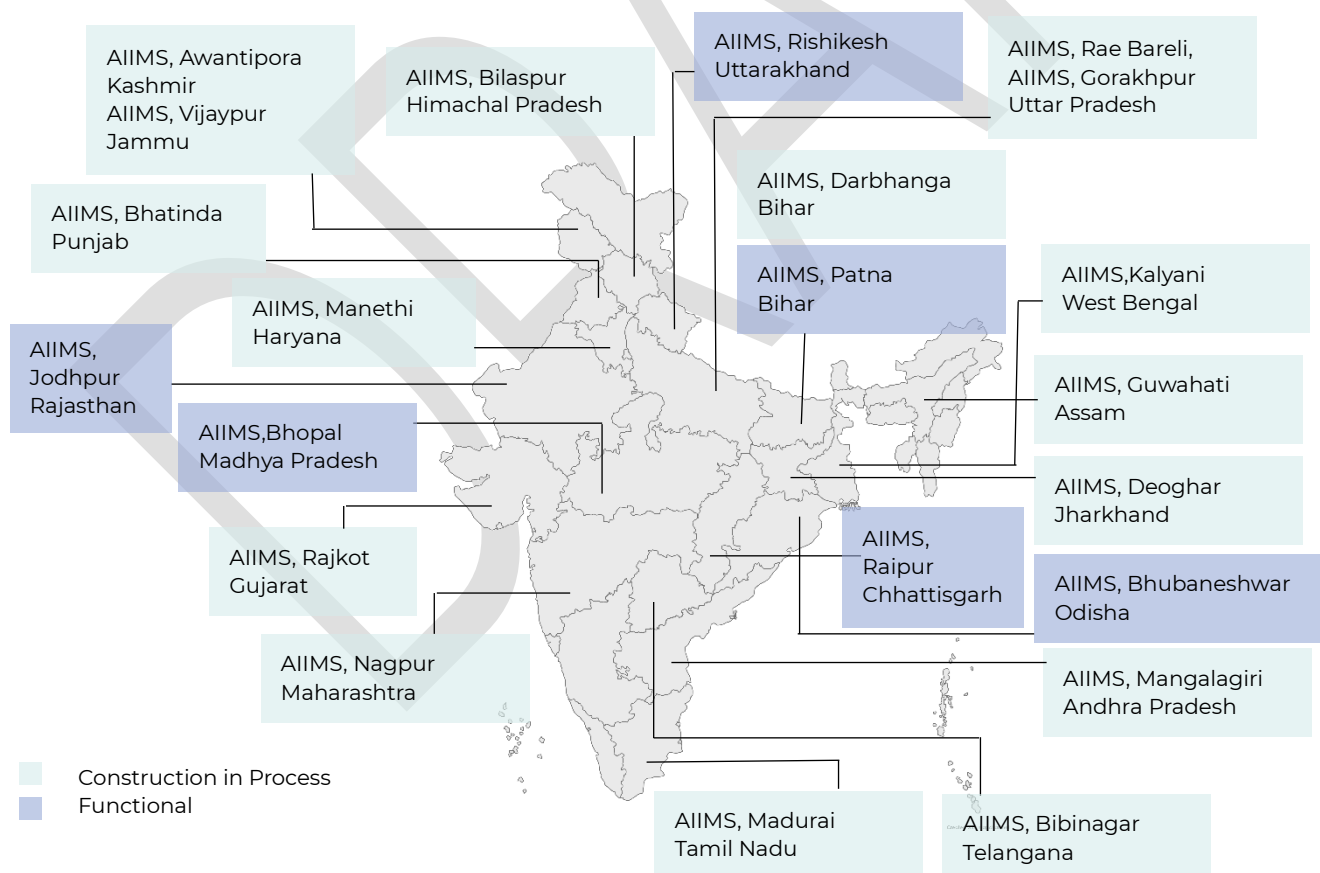
17. India Budget. (2023). *Demand for Grants. Ministry of Health and Family Welfare*.



Centrally Driven Initiatives in Medical Education

- Strengthening and upgrading existing state government/central government medical colleges to **increase MBBS and PG seats**.¹⁸
- Construction of **Super Specialty Blocks** in government medical colleges to enhance their capabilities.¹⁶
- **Approval of 22 new AIIMS** under Pradhan Mantri Swasthya Suraksha Yojana (PMSSY) Scheme, with 19 of them already offering undergraduate courses.¹⁶
- **Recognition of Diplomate of National Board (DNB) qualification** for faculty appointments, addressing the shortage of faculty.¹⁹
- **Enhancement of the age limit** for appointments/extension/re-employment in teaching positions in any medical college, allowing individuals up to 70 years to serve as teachers, deans, principals, or directors.²⁰
- The National Medical Council is planning to introduce the **National Exit Test (NExT)**, a common licentiate examination to replace the university-level final MBBS examination commencing from the next year May 2024.⁷
- Ministry of Health and Family Welfare (MoHFW) has launched a green field eHealth project, connecting medical colleges through **National Medical College Network (NMCN)** for e-Education and establishing a National Rural Telemedicine Network for e-Healthcare delivery.²¹
- In 2022, there were 50 medical colleges that had been shortlisted for the first phase of NMCN.²²

Figure 2 - New AIIMS set to launch across India²³



18. MoHFW. (n.d.). *Guidelines for Centrally Sponsored Scheme*. Department of Health and Family Welfare: Medical Education Division

19. MoHFW. (2017). *National Medical Commission Bill*. ME Division.

20. NMC. (2021). *National Medical Commission Draft Teachers' Eligibility and Minimum Qualifications in Medical Institutions Regulations*.

21. NMCM. (2023). *National Telemedicine Portal*. Retrieved 28 September 2023.

22. NMCM. (2023). *List of 50 Medical Colleges Shortlisted Under 1st Phase of NMCN*. NMCM.

23. MoHFW. (2018). *Annual Report (2018-2019): Central Medical Institutions and New AIIMS*.

Key Highlights for Other Streams/Branches of Medical Education



Dentistry

- Currently, India has a total of 329 dental colleges with 60 in the government sector and 269 in the private sector (as of October 2023).⁸
- Central Government has granted permission to increase seats in existing BDS Course at Govt. Dental Colleges/Institutions for the academic session 2019-20 for implementation of 10% reservation of seats for Economically Weaker Section (EWS) candidates in compliance with the Constitution (103rd Amendment) Act, 2019.²⁴
- The government has also initiated a Draft National Dental Commission Bill in the country to bring it at par with global standards by replacing the existing Dentists Act, of 1948.⁸



Nursing

- As of 2021, there were a total of 14,488 seats for B.Sc. nursing, 1,40,762 seats for Auxiliary Nursing and Midwifery (ANM), and 59,100 seats for General Nursing and Midwifery (GNM) across India.²⁵
- In 2023, National Nursing and Midwifery Commission Bill was launched for the regulation and maintenance of standards of education and services for nursing and midwifery professionals.²⁶



Paramedical

- In 2021, GoI had introduced National Commission for Allied and Healthcare Professions (NCAHP) Act, for regulating education and services of the allied and healthcare professional.¹⁰
- Establishment of the National Institute of Allied Health Sciences (NIAHS) and Regional Institutes of Allied Health Sciences (RIAHS) aims to address the growing demand for allied health professionals like laboratory technicians, assistants, dietitians, etc., and ensure high-quality education in this field.



Physiotherapy

- India has a total of 183 UG colleges for providing Bachelors in Physiotherapy (BPT) degree approved by Indian Association of Physiotherapists (IAP) (as of October 2023).²⁷
- Physiotherapists obtain licenses from state physiotherapy councils, which enforce a rigorous code of ethics and uphold high standards of continuing competency.²⁸

24. Dental Council of India. (2019). *Public Notice*.

25. National Health Mission. (2022). *Human Resources for Health in District Public Health System of India: State wise Report*. NHSRC.

26. PRSIndia. (2023). *The National Nursing and Midwifery Commission Bill, 2023*. Lok Sabha.

27. IAP. (2020). *List of Recognised / Approved Under-Graduate Colleges by Indian Association of Physiotherapist (IAP) In India*. Retrieved 28 September 2023.

28. MoHFW. (2017). *Model Curriculum Handbook Physiotherapy, Allied Health Section 2017*.



AYUSH

- To enhance the quality and operations of Ayurveda, Siddha, Unani, and Homoeopathy (ASU&H) Colleges throughout the country, the National Commission for Indian System of Medicine Act, 2020, and the National Commission for Homoeopathy Act, 2020 were introduced.¹²
- For 2022-23, the combined number of UG candidates in the Ayurveda, Siddha, Unani, and Homoeopathy streams were 13,274 and in the PG sector, the number of candidates were 5,566.¹²
- At present, 260 students from 32 countries are taking Ayush education in different institutes across India under the A Fellowship Scheme.¹²
- The Indian government's emphasis on integrating the Ayush system into the National Health ecosystem received a significant boost in the 2023-24 Budget. The budget allocation for the Ayush Ministry has been raised by 20 percent, totaling to INR 3,647 Crores.²⁹

B. Medical Research

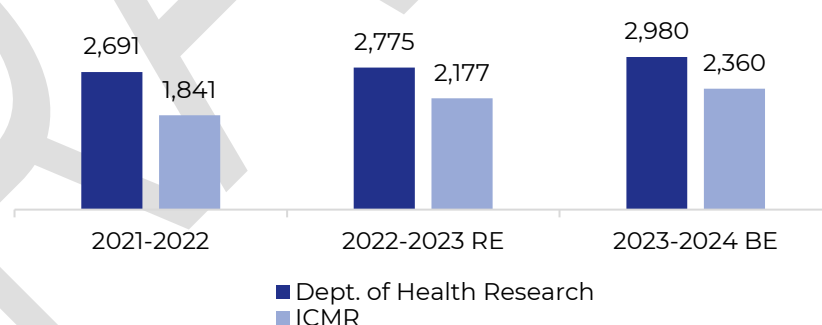
- Department of Health Research (DHR)** was created as a separate department within Ministry of Health and Family Welfare (MoHFW) to bring modern health technologies to the people through research and innovations related to diagnosis, treatment methods, and vaccines for prevention.
- In 2023-24, **about 3.3%** of MoHFW budget was allocated to Department of Health Research (INR 2,980 Crores).^{30,31}

Budget



For the 2023-2024, **~79%** budget of dept. of health research has been assigned to the Indian Council of Medical Research (ICMR), amounting to **INR 2,360 Crores.**³⁰

Figure 3 - Budget Allocation for Dept of Health Research (In INR Cr)³⁰



Indian Council of Medical Research (ICMR)³²

Under the Department of Health Research, ICMR holds a distinguished and influential position in advancing healthcare knowledge and research initiatives. It serves as a prominent knowledge centre, conducting vital research on diseases like malaria, tuberculosis, and AIDS. Their purview extends to nutrition, reproductive health, and targeted regional healthcare initiatives. Beyond research, ICMR significantly contributes by providing funding, training, and support for research projects nationwide. They facilitate the development of biomedical expertise through various schemes, fellowships, and training programmes, empowering a network of investigators across the country.

29. India Budget. (2023). *Demands for Grants, 2023-2024*, Ministry of AYUSH.

30. Department of Health Research. (2023). *Demands for Grants*, Ministry of Health and Family Welfare.

31. Department of Health Research. (2023). *Budget Reports*, Ministry of Health and Family Welfare.

32. ICMR. (2021). *Achievements*, Indian Council of Medical Research.

Major Activities of ICMR³²



*JRF: Junior Research Fellow, SRF : Senior Research Fellow, RA : Research Associate, STS: Short Term Studentship

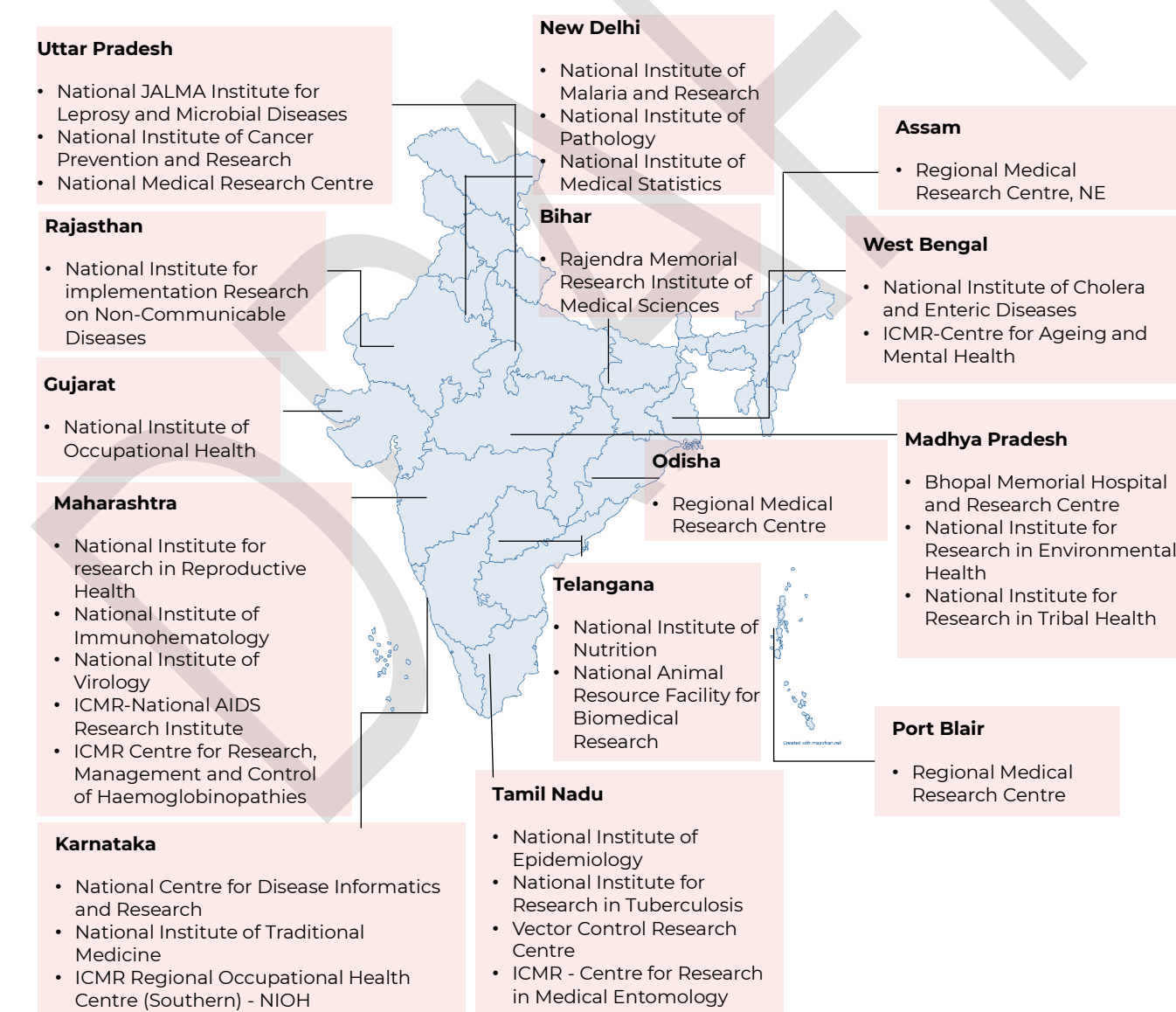
33. IAMRSN. (2019). *India's Antimicrobial Resistance Surveillance & Research Initiative*. Indian Council of Medical Research.

Strategic Framework of ICMR³⁴



Total Medical Research Institutes in India	 60 Number of Indian research institutions for biological and medical sciences³⁵	 31 Number of ICMR Institutions³⁶
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Figure 4 - Presence of ICMR Medical Research Institutes across India³⁷



34. ICMR. (2017). *ICMR Strategic Plan and Agenda*. Indian Council of Medical Research.

35. Dst.Gov. (n.d.). *List of Indian Institutions with Research Areas*.

36. ICMR. (2023). *Institutes*. Indian Council of Medical Research. Retrieved 03 October 2023.

37. ICMR. (2020). *Annual Report (2020-2021)*. Indian Council of Medical Research.

- **2022:** First mobile laboratory with **Biosafety Level-3 containment** was launched in Nashik, Maharashtra, marking a significant milestone for India.³⁸
- **2021:** Strengthening of existing **Viral Diagnostic and Research Labs (VDRLs)** and creation of four new **National Institutes of Virology (NIVs)** and a **National Institute for One Health** through ICMR.³⁹
- **2021:** Setting of **Regional Research Platform for WHO** South East Asia Region for research in emerging health threats.³⁹

Other Eminent Research Institutes

Jawaharlal Institute of Postgraduate Medical Education and Research (JIPMER)⁴⁰



- Focused on educational advancements, patient-centred research, and population health.
- The institute operates a tertiary care hospital that provides specialized medical care to patients from Puducherry and neighboring regions.
- JIPMER had collaborated with various international medical universities like: Heidelberg University, Deakin University School of Medicine, etc. to upscale public health strategies.
- For 2023-24, INR 1,490.43 Crores had been allocated by the Ministry of Health and Family Welfare for its expansion and development.¹⁷

Figure 5 – Jawaharlal Institute of Postgraduate Medical Education and Research



38. ICMR. (2021). *Annual Report Progress Report (2021-2022)*. National Institute of Virology.

39. ICMR. (2021). *E-Samvaad*. Indian Council of Medical Research.

40. JIPMER. (n.d.). *Home*. Retrieved on 03 October 2023.

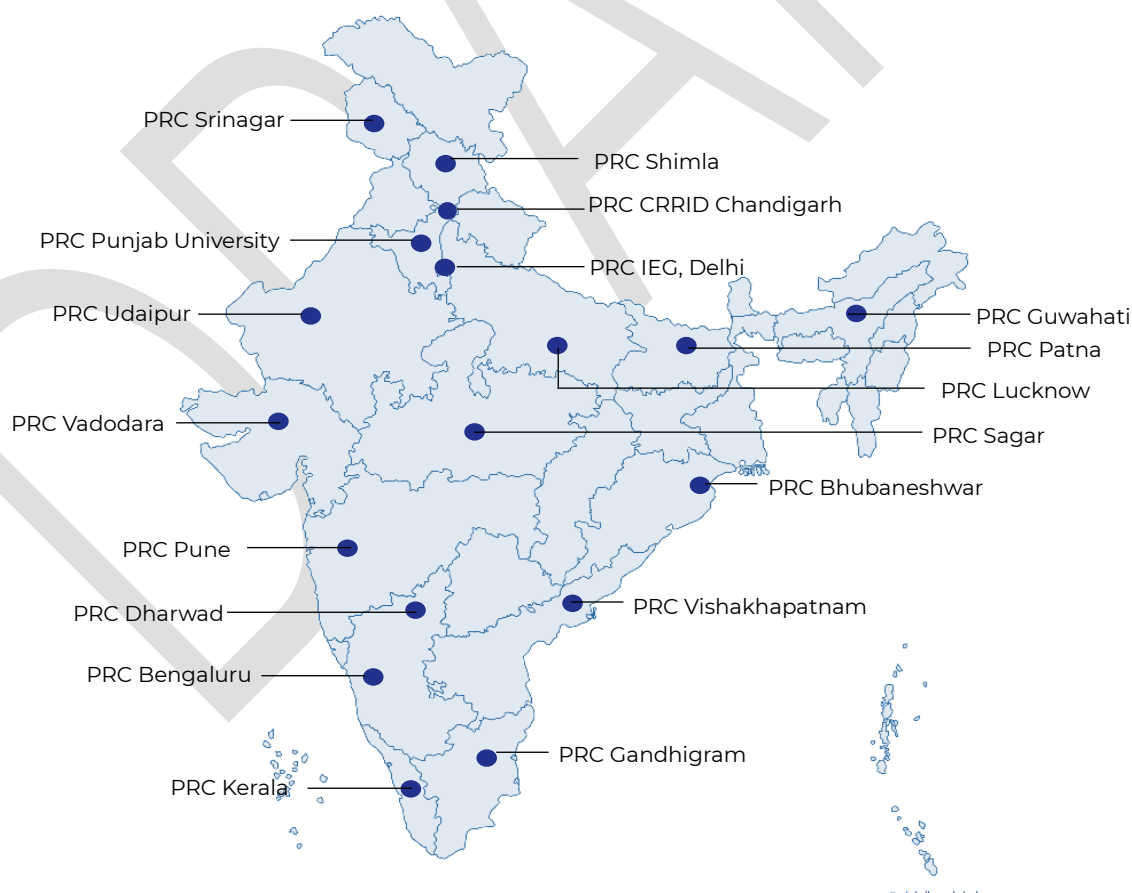
Population Research Institutes (PRC)^{41,42,43}



- PRCs were established to undertake research projects related to family planning, reproductive and child health, demographic research, population & development linkages, qualitative & quantitative aspects of population control, with a view to optimize the feedback from these research studies for strategies, formulation, implementation, and modifications of on-going schemes.
- Establishment of PRCs started with PRC, Delhi in 1957 and PRC, Kerala 1958 and currently, it has expanded to 18 PRCs, of which 12 are in universities while 6 are reputed research institutes (as of October 2023).
- They are provided with 100% central assistance in the form of grant-in-aid on a year-to-year basis.

Presence of Population Research Centres (PRC) across India

Figure 6 – Number of Population Research Centres in India



41. MoHFW. (2023). *Population Research Centre*. Retrieved on 03 October 2023.

42. MoHFW. (2021). *Research Institutions*. Retrieved on 03 October 2023.

43. MoHFW. (n.d.). *Population Research Centres*.

More National Institutes*¹⁷



- National Institute of Mental Health and Neuro Sciences, Bengaluru
- North Eastern Indira Gandhi Regional Institute of Health and Medical Sciences, Shillong
- Regional Institute of Medical Sciences, Imphal
- Regional Institute of Paramedical and Nursing Sciences, Aizawl
- Lokpriya Gopinath Bordoloi Regional Institute of Mental Health, Tezpur
- National Institute of Health and Family Welfare

**Kindly note that the institutes that have been mentioned in the Demands for Grants report of Ministry of Health and family welfare have only been taken in account for this section*

Opportunities and Challenges in Medical Education and Research

Opportunities



Introduction of New Curriculum and Courses

New courses and training programmes, bridge courses related to health research or public health and allied subjects are made available in the country addressing the evolving needs of the healthcare industry. This diversification allows medical students and professionals to explore different areas of interest, fostering an understanding of medical practice.

Technology Driven Learning and Healthcare Delivery

Medical students and professionals can access digital learning resources, interactive modules, and simulated patient cases to enhance their knowledge and skills. Additionally, telemedicine and webinars allow medical professionals to conduct remote lectures, workshops, and conferences, enabling continuous learning and collaboration within the medical community.

Collaboration with International Institutions

India presents a significant opportunity for research collaboration. By partnering with Indian institutions and researchers, organisations from around the world can leverage the country's expertise and resources to advance scientific knowledge and make significant contributions across various disciplines.

Addressing Specialist Shortage in Public Healthcare

Alternative models like district hospitals models aim to address the scarcity of medical specialists within publicly funded health systems. These models focus on employing specialized medical professionals strategically within district hospital with the aim of ensuring even remote or underserved areas have access to essential specialized medical services.

Higher Medical Education

PhD and post-doctoral medical education open avenues for specialized expertise, research advancements, and interdisciplinary collaboration, which is vital for addressing evolving healthcare needs in India. These programs will nurture skilled professionals, foster innovation and drive evidence-based practices to elevate healthcare delivery and outcomes.

44. Brahmapurkar KP, Zodpey SP, Sabde YD, Brahmapurkar VK. (2018). The need to focus on medical education in rural districts of India. National Medical Journal of India.
45. MoHFW. (n.d.). Rural Health Statistics 2020-2021. Statistics Division. Retrieved on 03 October 2023.



Limited Medical Colleges in Rural Areas

- There is an uneven distribution of medical colleges across India leading to a shortage of medical colleges in rural districts of India which leads to disparities in rural healthcare.^{44,45}
- None of the rural districts in Jharkhand, Jammu and Kashmir and Arunachal Pradesh have a medical college, further leading to regional disparities^{**}.⁴⁴
- Meanwhile, states such as Madhya Pradesh, Uttar Pradesh, Bihar, Assam, Punjab, Uttarakhand, Haryana and Rajasthan exhibit less than 30% of their rural districts equipped with medical colleges. In contrast, states like Andhra Pradesh and Telangana boast full coverage, with 100% of their rural districts housing medical colleges^{**}.⁴⁴

^{**} Data available as of 2018 study on medical education in rural districts of India

Shortage of Medical Faculty

- There is presently an acute deficit of medical faculty in Indian medical institutions, estimated at approximately 40%. The Medical Council of India's (MCI) guideline suggesting a 1:10 teacher-student ratio is not being upheld.⁴⁶
- In an attempt to address this issue, a few government institutions began appointing non-medical teachers to fill approximately 30% of the total departmental positions. However, this initiative faced substantial criticism from the medical community, as it resulted in decline in the overall quality of medical education.⁴⁶

Insufficient Clinical Research Activities

- Some emerging challenges in this sector encompass inadequate research and development investment, a fragile innovation ecosystem impeding idea translation into viable products, limited alignment of technology and trade, and a scarcity of full-time equivalent scientists per million people.⁴⁷
- The shortage of researchers in India poses a hurdle to the country's research and development expansion, presenting a significant challenge given factors like appealing research prospects abroad, limited incentives, and a lack of impactful research within India.⁴⁷

Shortage of Clinical Material

- Expensive and non-subsidized medical devices, coupled with inadequate medical facilities within medical colleges, lead to a situation where the patient load is low, and bed occupancy remains below 30%. This situation restricts the educational opportunities and experiences of students.⁴⁶
- Shortage of latest technology, lack of quality research activities leads to obstacles in backdated learning and optimal treatment, thus impacting the efficiency of medical education and research, in particular for rural and remote areas.⁴⁸

46. Charmode S. et al. (2020). Systemic Shortcomings in Medical Education System of India: A Review with Radical Solutions.

47. NITI Aayog. (2021). Reforms and Action Points to Strengthen Science, Technology and Innovation Ecosystem in India.

48. Basu J. (2022). Research on Disparities in Primary Health Care in Rural versus Urban Areas: Select Perspectives. *International journal of environmental research and public health*.

Cross-cutting Themes in Medical Education and Research



Gender

Significance: Promoting gender inclusivity in medical education and research is vital for a diverse, equitable healthcare workforce and improved health outcomes for all.

Current Efforts: The government encourages gender diversity in healthcare, industry initiates gender-sensitive research and policies to enhance gender balance in healthcare professions.

Way Forward: Implement comprehensive gender-sensitive curricula, encourage women's participation in research and leadership, and advocate for gender-inclusive healthcare policies and practices.



Boosting Infrastructure

Significance: Enhancing healthcare infrastructure is critical in providing quality medical education and facilitating advanced research, for ultimately improving healthcare delivery.

Current Efforts: The government has initiated reforms to increase the number of medical colleges and healthcare facilities, developing more specialized research institutes like Viral Research and Diagnostic Laboratories, Multi-Disciplinary Research units to boost healthcare infrastructure projects.

Way Forward: Continued investment in modern infrastructure, encourage public-private partnerships, and promote advanced research facilities to increase medical education and research capabilities.



Global Collaboration

Significance: Collaboration at the international level in medical education and research fosters knowledge exchange, innovations, and access to a broader talent pool. It can address global health challenges effectively.

Current Efforts: The Indian government is encouraging collaborations with foreign universities, like JIPMER collaborated with Heidelberg University and Deakin University School of Medicine to upscale public health strategies and participate in international research projects. Exchange programmes and joint research ventures are also growing.

Way Forward: Expanding more global collaborations through twinning programs, encouraging student and faculty exchanges, and facilitating joint research initiatives will create a more interconnected and knowledge-rich medical ecosystem in India.



Digital Technology

Significance: Digital technology is transforming medical education and research, offering enhanced learning resources, telemedicine, and advanced diagnostic tools. It holds the potential to revolutionize healthcare delivery.

Current Efforts: Government is promoting telemedicine and e-health portals like SASHAKT*, SAKSHAM*, TMIS* etc. to strengthen the public health sector's education, training, and human resources.

Way Forward: Foster digital literacy, advance telemedicine and health informatics education, and encourage research on AI applications in healthcare to fully harness the potential of digital technology in the medical domain.

*SASHAKT: Systematic Assessment of Healthcare Provider's Knowledge and Training

*SAKSHAM: Stimulating Advanced Knowledge for Sustainable Health Management

*TMIS: Training Management Information System

Ministries and Departments

Inter-ministerial convergence among ministries and departments is particularly crucial in the context of medical education and research. Various ministries and departments work together to formulate policies and implement programmes aimed at upskilling medical workforce and research scholars. These collaborative efforts are essential to efficiently address public health challenges.

Ministry	Department and Bodies	Functions and Roles
 स्वास्थ्य एवं परिवार कल्याण मंत्रालय MINISTRY OF HEALTH AND FAMILY WELFARE सत्यमेव जयते	 स्वास्थ्य एवं परिवार कल्याण मंत्रालय MINISTRY OF HEALTH & FAMILY WELFARE स्वास्थ्य एवं परिवार कल्याण विभाग DEPARTMENT OF HEALTH & FAMILY WELFARE Department of Health and Family Welfare	- Ensuring the availability and accessibility of quality healthcare services to all citizens through government hospitals, health centres, and other healthcare facilities. - It mainly functions in implementing health policies, family planning programmes, disease surveillance, Health Promotion and Education etc.
Ministry of Health and Family Welfare	 सत्यमेव जयते Department of Health Research	- Responsible for promoting and coordinating health research activities in the country. - Support and strengthen research in various areas of health and medicine to address public health challenges and improve healthcare services.
 विज्ञान एवं प्रौद्योगिकी मंत्रालय MINISTRY OF SCIENCE AND TECHNOLOGY सत्यमेव जयते	 Department of BioTechnology, Government of India Department of Biotechnology	- Plays a pivotal role in promoting and coordinating biotechnology-related activities in the country. - Involved in fostering the growth of biotechnology research, development of biotechnology applications in healthcare, governs industrial and agricultural biotechnology, along with providing support to bioinformatics and computational biology.
Ministry of Science and Technology	 विज्ञान एवं प्रौद्योगिकी विभाग DEPARTMENT OF SCIENCE & TECHNOLOGY Department of Science and Technology	- Plays a pivotal role in promoting scientific research, technology development and innovation across disciplines - Provides funds in research projects, develops scientific infrastructure, supports technology transfer and fosters collaboration with international partners to drive scientific advancements and address societal challenges through science and technology.



Ministry



Department and Bodies



Functions and Roles



MINISTRY OF
AYUSH

Ministry of AYUSH



National Commission for
Indian System of Medicine

**National Commission of
Homoeopathy (NCH)**



**National Commission for Indian
Systems of Medicine (NCISM)**



**Pharmacopoeia Commission
for Indian Medicine and
Homeopathy (PCIM&H)**

- NCH was established to oversee and regulate the homeopathic education and practice in the country.

- NCISM is responsible for setting standards for education, accreditation, and assessment of institutions offering education in Indian Systems of Medicine.

- PCIM&H acts as a Central Drug Testing cum Appellate Laboratory for Indian system of Medicine and Homoeopathy.



**Ministry of Human
Resource
Development**

- The Ministry of Human Resource Development is accountable for the progression of human resources in India.

Government Interventions in Medical Education and Research

A. Medical Education



Pradhan Mantri Swasthya Suraksha Yojana (PMSSY)

- The main objective of PMSSY is to address the shortage of human resources and educational capacity in the tertiary care system by establishing quality medical education, research and clinical care in the underserved area of the country.
- In 2023-24, the PMSSY has been allocated a budget of INR 3,365 Crores for the development of medical institutions and research facilities.
- Under this scheme there are two broad components on which the scheme mainly focuses on. They are :

Up-gradation of existing Government Medical Colleges/Institutions

- Up-gradation programme primarily aims at enhancing tertiary healthcare infrastructure by either constructing Super Specialty Blocks, Trauma Care Centres, etc., or acquiring medical equipment for established Government Medical Colleges or Institutions.⁴⁸
- 75 up-gradation Projects of existing Government Medical Colleges/Institutions have been approved under this scheme, out of which 60 have been completed.^{49,50}



Setting up of All India Institute of Medical Sciences (AIIMS)

- Establishment of 22 new AIIMS have been approved under this scheme, of which Six AIIMS approved under Phase - I are fully functional.⁴⁹
- An additional budget of INR 6,835 Crores has been allocated specifically for the establishment expenses of the new AIIMS under this scheme.³⁰



49. India Budget. (2023). *Expenditure Budget 2023-2024*. Ministry of Finance.

50. MoHEW. (n.d.). *Medical Education Policy & Medical Education: Annual Report 2015-2016*. MoHEW.

B. Research

ICMR Medical Innovation Fund



- An in-house scheme to promote creativity and innovation in biomedical research that supports novel ideas with scientific and medical implications.
- Each project is allocated a fund of INR 10,00,000 for idea testing.⁵¹
- Successful ideas with international patents may receive personal awards of INR 20,000 to one lakh.⁵¹

Establishment of VRDL*, MRU* and MRHRU* Scheme



- Through VRDL, MRU and MRHRU Scheme, DHR has built a strong ecosystem for health research by incorporating new technologies and treatments into public health.¹³
- The department has made significant progress in this scheme by sanctioning 145 VRDL, 100 MRUs and 30 MRHRUs in the 2022-23 period. (as of December 2022).¹³

ICMR Emeritus Scientist Scheme



- Launch of ICMR Emeritus Scientist Scheme was aimed at engaging retired scientist who have been actively engaged in biomedical research of a high standard.
- In the year 2022, five scientists were selected as ICMR Emeritus Scientists (IES).⁵³

Human Resource Development (HRD) Scheme



- A scheme to generate a pool of talented research scholars by upgrading skills of faculty of medical colleges, mid-career scientists etc. and encouraging them in taking up projects related to critical health and local health problems.¹²
- A total of 218 fellowships including 123 new fellowships of 2022-2023 have been supported under HRD Scheme.¹³

ICMR / DHR Policy on Biomedical innovation and Entrepreneurship



- A policy launched with the aim of promoting biomedical innovations across medical, dental, para-medical colleges/institutes, biomedical research, and academic institutions to encourage faculty, scientists to actively engage in innovations.
- This policy will ensure multi-disciplinary collaboration and promote start-up culture.⁵²

Grants in Aid (GIA) Scheme



- A scheme for inter-sectoral convergence and co-ordination for promotion and guidance on health research.¹³
- For the year 2022-2023, 148 ongoing projects have been funded along with 26 projects pertaining to Mental Health.¹³
- 938 Proposals have received response for the 2022-23 period.¹³

*VRDL: Viral Research and Diagnostic Laboratories
*MRU: Multi-Disciplinary Research Units
*MRHRU: Model Rural Health Research Units

51. ICMR. (2019). *Medical Innovation Fund*. Retrieved on 03 October 2023.

52. ICMR. (2023). *Launch Of "ICMR/DHR Policy On Biomedical Innovation & Entrepreneurship"*.

53. ICMR. (2020). *Emeritus Scheme*.

Way Forward

A. Medical Education



1. Standardisation



Standardisation of policies play a critical role in ensuring medical research is conducted ethically. Furthermore, it will also help in maintaining quality control in medical research and reduction in unauthorized medical institutes.

2. Strengthening of Workforce



In medical education, more focus will be on enhancement of quality of education and training through standardised and evidence-based teaching methodologies to have a greater number of skilled healthcare professionals. It is also critical to strengthen the workforce for preventive, promotive, rehabilitative, and palliative care aspects.

3. Upskilling



Professional development of medical professionals will now be accelerated through conferences, workshops, online courses, and knowledge-sharing platforms and this will continue to help in decreasing uneven distribution of skilled medical professionals in urban and rural areas.

B. Innovation and Research



4. More Research



Grants

Research grants facilitate scientific advancement by providing more financial support to researchers and institutions. Focus can be on providing adequate research grants to attract and retain talented and young researchers within the country.

5. Collaboration



ICMR Institutes will be making greater efforts for conducting collaborative projects with medical colleges in order to mentor research culture among their faculty. Alongside, focus is now on linking national AMRSN to public health interventions in a dynamic mode.

6. Research in Rare Diseases



Clinical research studies can be more focused on the field of rare or genetic diseases to strengthen national health databases like Clinical Trials Registry-India (CTRI) to draw programmatic implications.

Best Practices

1. Innovations in rolling out certificate course in community health, Uttar Pradesh⁵⁴

Background: To convert sub-centres, PHCs and UPHCs into Health and Wellness centres in Uttar Pradesh, requires incremental addition of human resources in the form of Community Health Officers. With this aim, a conceptual framework for rolling out certificate courses in Community Health in order to create a training eco-system to operationalise programme study centres and to develop a pool of trainers and structured implementation of academic activities.

Steps taken by Government:

- Generated excitement by involving stakeholders in state-level orientation gatherings.
- Established a training ecosystem by initiating operationalisation of Programme Study Centres (PSCs) with Training of Trainers (ToTs) at PSC level, aiming to cultivate a proficient pool of educators and ensuring structured execution of academic engagements.
- Ensuring quality through standardised teaching materials, academic planning tools, and mentorship visits.

Figure 7 – Classroom teaching for certification course in Community Health



Impact Generated:

- Successfully established and operationalised 32 Programme Study Centres.
- Established a resource pool of 339 academic counsellors.
- Developed 71 standard presentations to aid academic counsellors in conducting well-structured theory and practical classes.
- Played a key role in implementing mid-course corrections and ensuring training quality.

54. NHSRC. (2017). National Summits on Good, Replicable Practices & innovations in Public Healthcare Systems in India

2. Training of tribal girls as ANM and GNM in two tribal predominant districts of Chhattisgarh - Jan Swasthya Sahyog (JSS)⁵⁵

Background:

The district of Bilaspur is home to a population of 2.6 million, with 1.2 million being part of the tribal community. The area has limited access to healthcare services. Recruiting healthcare professionals like nurses, paramedics (lab technicians, OT technicians, pharmacists), and doctors willing to work in these regions seemed challenging. Additionally, a cultural and language barrier exists for tribal patients seeking healthcare. Hence this programme was launched to address these challenges by training a substantial number of tribal girls as Auxiliary Nurse Midwives (ANM) or General Nursing Midwives (GNM) to serve in their local communities and hospitals.

Steps taken by Government:

- In 2010, JSS established a Nursing school in collaboration with the tribal welfare department of the Government of Chhattisgarh. This institution received recognition from esteemed bodies such as the Indian Nursing Council, State Nursing Council, and the Chhattisgarh health department. The school was granted permission to admit 25 candidates annually for both the ANM and GNM courses.
- The selected candidates for these courses were from families residing below the poverty line, primarily belonging to the tribal or Dalit communities. The selection process commenced at the village level, involving community visits and outreach to family members and tribal girl hostels to disseminate information about the programme. Selection criteria included evaluating the candidate's merit, social background, social sensitivity, and awareness regarding rural and tribal issues.
- These trainees underwent comprehensive exposure throughout all tiers of the programme structure, enhancing their knowledge and skills to effectively serve their communities.

Figure 8 – Training for ANM and GNM in rural communities



Impact generated:

The training programme started in 2010. To date, two batches of ANMs have successfully completed the programme, achieving a perfect 100% pass rate. Notably, 90% of the students received distinctions. During their internships, these individuals honed their skills across multiple specialties such as emergency care, neonatal care, obstetrics, and community health. Following their training, 60% of these ANMs were placed in the Government sector, serving in sub-centres, primary health centres (PHCs), and community health centres (CHCs). Another 30% found employment in the voluntary sector, and the remaining 10% secured positions in the private sector.

55. NHSRC. (2015). *Best Practices in Area of Tribal Health* by NHSRC India.

3. Harnessing the potential of India's medical colleges to improve maternal nutrition services (Uttar Pradesh, Bihar and Jharkhand)⁵⁶

Background:

An initiative to enhance the integration of the Maternal Infant Young Child Nutrition (MIYCN) course within specific government medical colleges and hospitals in Uttar Pradesh, Bihar, and Jharkhand. Efforts were made to embed it comprehensively within the undergraduate curriculum, service provision, research activities, and government programme support. This initiative aims to produce a generation of well-trained medical practitioners, both in the public and private sectors, who possess a solid foundation in maternal nutrition.

Steps taken by Government:

- **Establishing Commitment and Ownership:** State government and medical college leaders demonstrated commitment and ownership by formalizing their involvement through Memorandums of Understanding (MoUs).
- **Visioning Exercise:** Collaborated with medical college departments of Paediatrics, Obstetrics and Gynaecology (OBGY), and Community Medicine to conduct a visioning exercise, setting the project's direction.
- **Gap Assessment:** Conducted a comprehensive gap assessment within the selected medical colleges, evaluating MIYCN knowledge, skills, teaching-learning methods, research priorities, and practices. This assessment informed the development of a strategic roadmap and implementation plan.
- **Formation of Expert Groups and Committees:** Established expert groups at the national and state levels, along with interdepartmental committees comprising senior academicians, practitioners, and researchers. These groups provided guidance and oversight to ensure the successful implementation of the initiative.
- **Capacity Building:** Conducted capacity building initiatives for faculty and hospital staff, including postgraduate residents and nursing staff from the relevant departments. The training focused on adopting the strengthened MIYCN curriculum, service delivery protocols, and a Quality Improvement (QI) approach.

Impact generated:

- Early initiation of breastfeeding in uncomplicated C-section deliveries surged from 0% to an impressive 89% within a four-month period.
- Improved patient flow and management resulted in nearly three-fourths of the patients receiving comprehensive Antenatal Care (ANC) services during the specified period.
- Around 84% of antenatal mothers underwent weight, haemoglobin, and blood pressure measurements, ensuring a thorough health evaluation.
- Maternal nutrition counselling was provided to 76% of women, empowering them with crucial information.
- At Government Medical College, counselling on exclusive breastfeeding in the postnatal ward saw a remarkable increase from 13% to an impressive 94% within a nine-week timeframe.
- Substantial training efforts were undertaken, with approximately 600 doctors, nurses, and counsellors in Bihar and 750 in Uttar Pradesh receiving training on service delivery protocols and the Quality Improvement (QI) approach.

56. NITI Aayog. (n.d.). NITI Aayog's Health & Nutrition Insights: Vol-II.

Recommended Readings

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2. Ministry of Health and Family Welfare. (2020). *Governance Reforms in Medical Education (2014-2020)*. Ministry of Health and Family Welfare. <https://main.mohfw.gov.in/sites/default/files/Education%20Booklet1.pdf>
3. NITI Aayog. (2023). *Annual Report 2022-23*. NITI Aayog. https://www.niti.gov.in/sites/default/files/2023-02/Annual-Report-2022-2023-English_1.pdf
4. Indian Council of Medical Research. (n.d.). *Grand challenges/broad goals as relevant to the nation to be achieved in XII plan*. https://main.icmr.nic.in/sites/default/files/reports/14_0.pdf
5. World Health Organization. (2012). *Health education: theoretical concepts, effective strategies and core competencies*. WHO. https://applications.emro.who.int/dsaf/EMRPUB_2012_EN_1362.pdf
6. Sitthi-Amorn, C., & Somrongsong, R. (2000). Strengthening health research capacity in developing countries: a critical element for achieving health equity. *BMJ (Clinical research ed.)*, 321(7264), 813–817. <https://doi.org/10.1136/bmj.321.7264.813>
7. Council Reports. (2021). *Report 3 of the council on medical education- Rural Health Physician Workforce Disparities*. American Medical Association. https://councilreports.ama-assn.org/councilreports/downloadreport?uri=/councilreports/n21_cme_03_annotated.pdf
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